

Kevin Jacob

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EDUCATION

University of Colorado Denver

Master of Science, Computer Science

Expected May 2026

University of Colorado Boulder

Bachelor of Science, Computer Science

May 2024

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, TypeScript, JavaScript, SQL, Scala, C
Frameworks & Libraries: Spring Boot, FastAPI, Django, Node.js, LangChain, Numpy, Pandas, Scikit-learn, TensorFlow, HuggingFace, Axios
Cloud & DevOps: AWS (Lambda, S3), Docker, Git, CI/CD, Agile, Jira, Postman
Databases: MongoDB, MySQL, Cassandra, SQL
Certifications: AWS Certified AI Practitioner (2025)

EXPERIENCE

Emergetech Business Solutions Inc.

Software Developer Intern

June 2024 - August 2024

- Improved customer response quality by integrating the OpenAI API into a call center platform to generate AI-powered real-time scripts, reducing agent handle time by enabling automated prompt-driven assistance across a full-stack Spring Boot and React application.
- Enhanced management reporting capabilities by building a dynamic SQL query builder that filtered and aggregated records across large datasets using tag-based logic, reducing manual data analysis time through automated query generation.

Expedia Group

Capstone Member

August 2023 - April 2024

- Accelerated listing quality improvement for VRBO property managers by building an end-to-end AI-powered recommendation platform, delivering automated scoring across hundreds of listings by combining web scraping, the OpenAI API, and LangChain prompt engineering deployed on AWS Lambda.
- Improved recommendation accuracy by engineering and iteratively optimizing GPT prompts for structured score extraction, reducing inconsistent model outputs through systematic evaluation and refinement of the LangChain pipeline.

OptIn Inc.

Applications Developer Intern

June 2023 - August 2023

- Reduced scheduling conflicts for an employee portal by building a constraint-based predictive scheduling engine in Python and Django using Google OR-Tools, automating shift assignments across hundreds of workers by modeling labor demands and availability as an optimization problem.
- Improved algorithm reliability by leading end-to-end testing and constraint validation, catching edge case failures through systematic unit testing before production deployment.

University of Colorado

Teaching Assistant - Algorithms, Database Systems, Intro to Programming

January 2023 - Present

- Improved student comprehension across three courses by conducting weekly office hours for 100+ students, accelerating understanding of complex algorithms and data structures by debugging code in real time and translating technical concepts into accessible explanations.

PROJECTS

F1 Pit Stop Strategy Optimizer | Python, PyTorch, GRU, Scikit-learn, FastF1 API

- Achieved 52.5% pit stop strategy accuracy across 122 race-driver combinations from the 2024 F1 season by designing a compound-specific GRU deep learning system trained on 28,000+ lap sequences from 2022-2023 race data, representing a +21.5% improvement over the brute-force baseline.
- Improved tire degradation prediction accuracy by 40% over the moving average baseline by engineering a lap-to-lap delta prediction target, building a full data pipeline on the FastF1 API with per-race MinMax normalization, and conducting five ablation experiments across nine model configurations to identify the optimal architecture.

AI Cybersecurity Defender | Python, FastAPI, GPT-4o, LangChain, CICIDS2017

- Achieved 94% network threat detection accuracy by designing a production-grade AI agent using GPT-4o and FastAPI, combining a rule-based classification engine with an LLM-driven reasoning pipeline to detect intrusions in real time across the CICIDS2017 benchmark dataset.

Lyric Analyzer | Python, Scikit-learn, TensorFlow, HuggingFace, Spotify API

- Improved song classification accuracy by building a full ML pipeline from data ingestion to deployment, training an SVM classifier with TF-IDF feature extraction on Spotify API data and fine-tuning a feedforward neural network for generative lyric extension across genre-specific corpora.

Conference Room Booking Application | Java, Spring Boot, Thymeleaf, MySQL, Bootstrap

- Improved booking reliability for a multi-user scheduling system by building a full-stack Java/Spring Boot application, implementing OOP design patterns (Decorator, Factory, Singleton, Observer) and owning the MySQL database repository layer for user, room, and booking data.

Music Database | Node.js, MySQL, Axios, Mocha, Chai

- Increased data retrieval reliability for a full-stack music search application by integrating the AudioDB API via Axios and implementing comprehensive unit and integration tests through Mocha and Chai, validating correctness of all API responses and database queries.

RELEVANT COURSEWORK

Machine Learning, Natural Language Processing, Algorithms, Database Systems, Operating Systems, Artificial Intelligence, Advanced Data Science, Cybersecurity, Object-Oriented Analysis and Design, Human Computer Interaction